GGG-R-96E
June 6, 1978
SUPERSEDING
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May 27, 1968

FEDERAL SPECIFICATION

RAKE, HAND (GARDEN AND LAWN)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers garden rakes suitable for raking, and smoothing soil, and lawn rakes suitable for raking grass clippings, leaves, twigs, and litter from lawns.

1.2 Classification.

1.2.1 Types, classes, styles, and sizes. Hand rakes shall be of the following types, classes, styles, and sizes as specified (see 6.1).

Type II - Garden.

Class A - Bow.

Size 1.
Size 2.

Class B - Level head.

Size 1.
Size 2.

Type III - Lawn.

Class A - Steel, flexible head.

Style 1 - Full radial head.

Size 1.
Size 2.

Style 2 - Semiradial head.

Size 1.
Size 2.

Class B - Wood.

Class C - Lightweight metal straighthead.

Class D - Lightweight metal, adjustable head.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of rakes which are otherwise in accordance with the requirements of this specification.

3.2 Material. The quality of the materials not hereinafter specified shall be in accordance with vendor's standard commercial practice.

3.3 Measurements. All dimensions shall have a tolerance of ± 1/8 inch unless otherwise specified. Rake thickness is measured across the top or head of the rake. Tooth depth is measured from the tip of the tooth to the upper edge of the rake head. Width is measured across the teeth to the two outside edges. Handle length is measured from end to end including the ferrule.

3.4 Hardness of metal teeth and tines. Metal teeth (type II, classes A and B rakes) shall be of sufficient hardness to withstand dropping and loading tests (see 4.4.1 and 4.4.3).

3.5 Teeth and tines. The teeth and tines shall be of commercial length, width, and uniformly spaced and shaped.

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3.5.1 Shanks. The shank shall be centered between ends of the head, plus or minus 1/8 inch, for proper balance. The shank and head shall not fracture or crack, and the permanent set shall not exceed 1 inch when loaded (see 4.4.2).

3.6 Handles.

3.6.1 Wood handles. Wood handles shall be fabricated from hardwood in accordance with the vendors standard commercial practice and free from all defects that would affect the serviceability. When tested as described in 4.4.1, 4.4.2, and 4.4.4, the handles shall not have or loosen.

3.6.2 Metal handles. Metal handles shall be made of lightweight tubular metal not less than 1 inch in diameter and shall be furnished with a cushion-type hand-grip not less than 4 inches in length. The hand-grip shall be permanently affixed to the extreme end of the handle. Rake handles for class C rakes shall be of two sections with no one section over 48 inches long. When assembled, the class C and D rakes shall be tested in accordance with 4.4.4. The handles shall not separate or loosen.

3.7 Ferrules and caps. Ferrules shall be of steel, seamless or the seam shall be closed by a continuous weld. Caps when used shall be cupped or the end of the ferrule may be rounded to form a cap.

3.8 Nails and rivets. Nails and rivets for attachment of handle or ferrule, shall be steel, oval or round head.

3.9 Finish. Heads, ferrules, and all exposed parts of all rakes shall be finished in accordance with manufacturer's commercial practice.

3.10 Identification marking. The head, handle or the brace shall be branded or marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.11 Type II, garden rakes. Type II garden rakes are commercially designed for raking and smoothing soil. Teeth and other portions of the head shall not flatten, chip, crack, deform or fracture when tested in accordance with 4.4.1. No loosening of the handle shall occur. When tested in accordance with 4.4.3, no tooth shall take a noticeable permanent set after supporting for 2 minutes the tooth test load specified for the rake in the applicable table.

3.11.1 Ferrule. A steel tubular ferrule of not less than 3-3/4 inches in length shall be provide. The ferrule shall be tapered and fastened to the handle with either epoxy or at least one nail or rivet which need not extend entirely through the handle but shall pass entirely through the shank. If the nail or rivet should penetrate the opposite side of the ferrule, it shall be neatly headed. The ferrule shall be provided with a cap as specified in 3.7.

3.11.2 Class A, bow.

3.11.2.1 Head and bow. The head and bow shall be forged in one piece from bar steel. The bow shall be of circular cross section and at least 9/32 inch in diameter. The tang ends of the bow shall be forced into the handle and reinforced with a ferrule. The tang shall be secured to the handle with either epoxy or at least one nail or rivet. The bow construction shall not prevent the top of the head from being used for leveling operations.

3.11.2.2 Teeth. The teeth shall be as shown on figure 1.

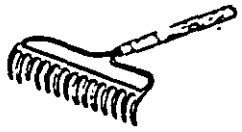


FIGURE 1. Type II, class A, garden rake, bow.

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3.11.2.3 Type II, class A rakes shall be similar to figure 1 and shall conform to table I for the size specified (see 6.1).

TABLE I. Type II, class A garden rake, bow

Size	Head				Handle	Test loads	
	Teeth Number	Depth (Min.)	Thickness ± 1/32	Width (Min.)	Length (Min.)	On tooth (4.4.3)	On handle (4.4.4)
		Inches	Inch	Inches	Inches	Pounds	Pounds
1	14 or 15	3	1/4	13-1/2	59	75	300
2	16	3	1/4	16	59	75	300

3.11.3 Class B, level head.

3.11.3.1 Head. The head and shank shall be forged in one piece from bar steel.

3.11.3.2 Teeth. The teeth shall be as shown on figure 2.

3.11.3.3 Shank. The rake shall have a single central shank which shall be forced into the handle and reinforced with a ferrule. The shank shall be so arranged as not to interfere with the use of the top of the head for leveling operations. When tested in accordance with 4.4.2 the shank and head shall not fracture or crack, and the permanent set shall not exceed 1 inch. No loosening of the handle shall occur.

3.11.3.4 Type II, class B rake shall be similar to figure 2 and shall conform to table II for the size specified (see 6.1).

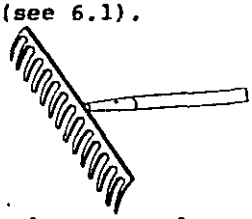


FIGURE 2. Type II, class B garden rake, level head.

TABLE II. Type II, class B garden rake, level head

Size	Head				Handle		Test loads	
	Teeth Number	Depth (Min.)	Thickness + 1/16	Width (Min.)	Length (Min.)	On tooth (4.4.3)	On handle (4.4.4)	On shank (4.4.2)
		Inches	Inch	Inches	Inches	Pounds	Pounds	Pounds
1	14	3	5/16	14	65	75	300	16
2	16	3	5/16	16	65	75	300	16

3.12 Type III, lawn rakes. Lawn rakes shall be suitable for raking grass clippings, leaves, twigs, and litter from lawns without injury to the grass or roots.

3.12.1 Class A, steel, flexible head. Class A flexible head rakes shall consist of strips of flat spring steel with bend ends forming the head and tines, a handle and a ferrule or socket suitable for securing the head to the handle.

3.12.1.1 Handle fastening. A ferrule or socket shall be provided, having a hole suitable for securing the handle by means of a common rivet, screw or bolt with nut. The ferrule or socket shall be arranged for a tight or drive fit with the handle. If any backing spring is employed on the rake, fastenings shall be supplied.

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3.12.1.2 Flexibility. The tines shall be flexible and the permanent set shall not exceed 1/8 inch when tested in accordance with 4.4.5.

3.12.1.3 Class A, style 1, full radial head. The tines forming the head shall be separated at the raking end, grouped together and securely fastened to a steel ferrule at the handle end. The ferrule shall extend at least 2-1/4 inches over the handle. The head of the rake shall have a steel spreader bar attached to the tines to keep them evenly separated and to hold the tines in a fixed position at all times. The spreader bar shall be located between the tips of the tines and the ferrule. Style 1 rakes shall be similar to figure 3 and shall conform to table III for the size specified (see 6.1).



FIGURE 3. Type III, class A, style 1 lawn rake, steel, flexible full radial head.

TABLE III. Type III, class A, style 1 lawn rake, steel, flexible full radial head

	Head		Handle
	Width	Tines	Length
Size	(Min.)	Number (Min.)	(Min.)
	Inches		Inches
1	18	18	42
2	23	24	57

3.12.1.4 Class A, style 2, semiradial head. The tines forming the head shall be separated at the raking end, grouped together and securely fastened to a steel ferrule at the handle end. The ferrule shall extend at least 2-3/4 inches over the handle. The head of the rake shall have a steel spreader bar attached to the tines to keep them evenly separated and to hold the tines in a fixed position at all times. The spreader bar shall be located between the tips of the tines and the ferrule. A backing spring or similar device to provide constant pressure on the head shall be secured to the ferrule, with the spring prongs attached to the spreader bar, to relieve the pressure at the stress point on the tines. Class A, style 2 rakes shall be similar to figure 4 and shall conform to table IV for the size specified (see 6.1).



FIGURE 4. Type III, class A, style 2 lawn rake, steel, flexible semi-radial head.

TABLE IV. Type III, class A, style 2, lawn rake, steel, flexible, semi-radial head

	Head		Handle
	Width	Tines	Length
Size	(Min.)	Number (Min.)	(Min.)
	Inches		Inches
1	18	18	52
2	21	22	52

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3.12.2 Class B, wood. Class B, wood rakes shall consist of a head and teeth of wood, a handle and a ferrule and metal braces.

3.12.2.1 Head. The head and teeth shall be sound, straight-grained, thoroughly seasoned hardwood free from knots in accordance with the vendor's standard commercial practice.

3.12.2.2 Teeth. The teeth shall be straight, circular in cross section and tapered toward the free ends. The teeth shall be forced or tightly fitted into the top of the head and secured with metal pins or nails driven through the teeth and top of head.

3.12.2.3 Ferrule. A ferrule shall be mortised into the head and fastened thereto with one or more screws. The ferrule shall be not less than 12 inches in overall length. When the rake is supplied assembled the ferrule shall be secured to the handle by bolting, by riveting, or two or more nails. When the rake is supplied knocked down, the handle and the ferrule shall be so constructed that when assembled they shall not loosen, turn, slip, or become detached from each other when tested in accordance with 4.4.4.

3.12.2.4 Braces. The head shall be braced against the handle with at least three circular bows of steel tubing or solid aluminum rod, not less than 3/16 inch in diameter. The bows shall pass through the ferrule, and the ends of such bows shall be secured to the head with nails or screws.

3.12.2.5 Type III, class B rakes shall be similar to figure 5 and shall conform to table V (see 6.1).

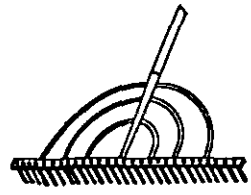


FIGURE 5. Type III, class B lawn rake, wood.

TABLE V. Type III, class B, lawn rake, wood

Head				Handle		
Teeth		Depth	Thickness + 1/16	Width	Length	Test load on handle (4.4.4)
Number	Diameter (Min.) 1/	(Min.)	(Min.)	(Min.)	(Min.)	
	Inch	Inches	Inch	Inches	Inches	Pounds
28	3/8	4-1/4	3/4	26	60	100

1/ Measured at head. (+ 1/32 inch tolerance)

3.12.3 Class C, straight head. Class C, straight head rakes shall be of a design that can readily be assembled and disassembled with a standard screwdriver and an adjustable wrench or equal. The rake shall have a rigid back fabricated from channeled lightweight metal with a flange or other device secured to the back for attaching the handle. The tines shall be fabricated from strips of spring steel and secured to the back of the rake in such a manner as to hold the tines in position at all times. Class C rakes shall be similar to figure 7 and shall conform to table VI for the size specified (see 6.1).

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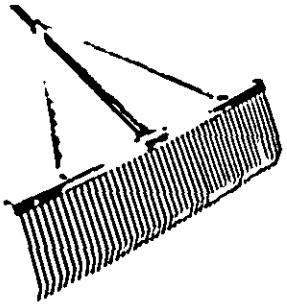


FIGURE 6. Type III, class C, garden rake, lightweight metal, straight head.

TABLE VI. Type III, class C, garden rake, lightweight metal, straight head

Head		Handle	
Width	Tines	Length	Test load on handle
(Min.)	Number (Min.)	(Min.)	(4.4.4.)
Inches		Inches	Pounds
29	47	60	100

3.12.3.1 Flexibility. The tines shall be flexible and the permanent set shall not exceed 1/8 inch when tested in accordance with 4.4.5.

3.12.4 Class D, adjustable head. Class D rakes shall be of a design that can be readily adjusted from a full radial flexible head to a minimal radial rigid head similar to figure 7 and all positions in between. Tines shall be attached to a movable carrier at the handle end and move freely through a steel spreader bar keeping them equally spaced in all positions. Adjustment shall be accomplished without the use of tools and in such a manner that the selected adjustment will be maintained in use.

3.12.4.1 Flexibility. The tines shall be flexible and permanent set shall not exceed 1/8 inch when tested in accordance with 4.4.5.

3.12.4.2 Handle. The handle shall be of lightweight metal fitted with a suitable hand grip with provision for hanging storage (see 3.6.2).

3.12.4.3 Type III, class D, lawn rakes shall be similar to figure 7 and shall conform to table VII.

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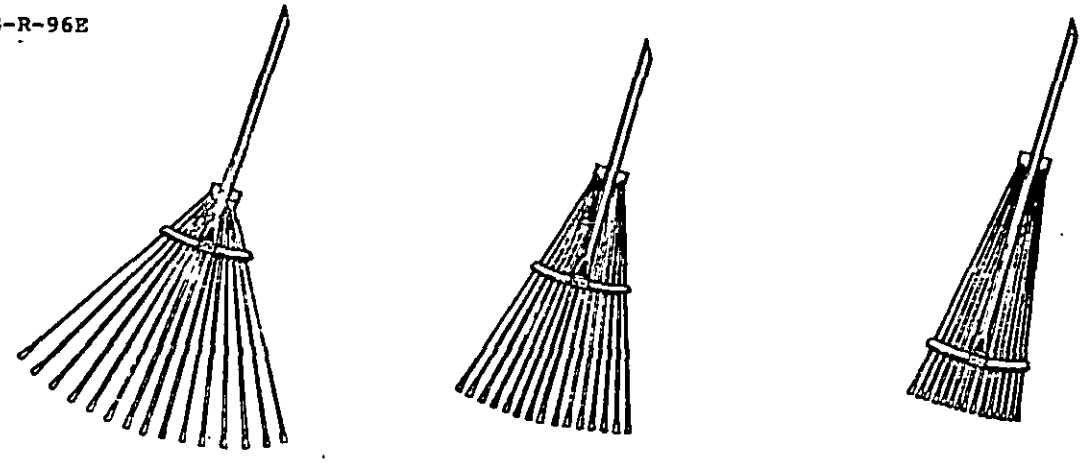


FIGURE 7. Type III, class D. lawn rake, lightweight metal, adjustable head.

TABLE VII. Type III, class D, lawn rake, lightweight metal, adjustable head

Head			Handle	
Width		Tines	Length	Test load on handle (4.4.4)
Adjustment Range (Min.)	(Max)	Number (Min.)	(Min.)	
Inches	Inches		Inches	Pounds
7	25	14	48	100

3.13 Workmanship. The workmanship shall be of the highest grade throughout, and equal in every respect to good commercial practice. Rakes shall be free from any imperfections that may affect their appearance, durability, or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Sampling for visual and dimensional inspection. A random sample of hand rakes shall be taken from each lot offered for inspection of visual and dimensional characteristics in accordance with MIL-STD-105. Inspection level shall be I. The AQL shall be 1.5 percent defective.

4.2.1 Sampling for tests. A random sample of rakes selected in accordance with MIL-STD-105, shall be taken from each lot for tests specified in 4.4. The inspection level shall be S-4. The AQL shall be 1.5 percent defective.

4.3 Visual and dimensional examination. Each of the sample rakes taken shall be examined for material, dimensions, protective coating, marking, finish, workmanship, and general compliance with this specification. Any rake in the sample containing one or more defects shall be rejected, and if the number of defective rakes in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 Tests.

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4.4.1 Drop test, type II rakes. The head is dropped teeth foremost, a distance of 4 feet, three times, on a flat, horizontal steel die block. In dropping the head, the free end of the handle shall be loosely pivoted at the level of the surface of the die block.

4.4.2 Shank test, type II rakes. The full width of the head and as much of the depth of the head as is possible shall be rigidly clamped or secured in a stationary fixture with the teeth pointed upward and the handle horizontal. The shank test load specified in table II shall be applied vertically downward to the handle for 120 to 125 seconds at a distance of 36 to 37 inches from the point where the shank joins the head. The vertical distance from the point of application of the load to any flat horizontal surface shall be measured before the load is applied and after it is removed, and the difference between these distances shall be the set.

4.4.3 Tooth strength test, type II rakes. The head of the rake shall be secured or supported in a stationary fixture and the load taken from the applicable table shall be applied in the center of the tooth in a direction opposite to the handle for period of 2 minutes. Immediately examine the tooth after removal of the test load. Any evidence of permanent set shall constitute failure.

4.4.4 Handle pulling test, types II and III, class B, C and D rakes. The head shall be clamped or secured and the handle test load, specified for the rake in the applicable table, shall be applied for 2 minutes as a pulling load on the handle.

4.4.5 Flexibility of tines test for type III, classes A, C, and D rakes. With the plane of the head horizontal and the tines pointed upward, a weight of 3 pounds suspended from any two adjacent teeth applied 1 to 2 inches in from the tip ends shall move the points of the teeth at least 1-1/2 inches, and upon removal of the weight after 2 minutes, the permanent set of the tines shall not exceed 1/8 inch. The set shall be the vertical distance between the top of the tested tines and adjacent tines not tested.

4.5 Examination of preparation for delivery. An examination shall be made to determine whether the packing complies with the requirements of Section 5. A random sample of shipping containers or bundles shall be taken from each lot offered for inspection in accordance with MIL-STD-105. The sample unit shall be one shipping container or one bundle fully prepared for delivery. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

5. PREPARATION FOR DELIVERY

5.1 Commerical packing. Each rake shall be cleaned and preserved in accordance with commercial practice. The rakes shall be packed in fiberboard boxes or strongly strapped bundles that will insure safe delivery at destination, provide for safe redistribution by the initial receiving activity, and be acceptable by common carrier under the Uniform Freight Classification and National Motor Freight Classification.

5.2 Marking. Marking shall be as specified in the contract or order.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, style, and size (if applicable) (see 1.2).

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